Date:

Paper 2017

Relative Efficiency

Relative efficiency refers to the comparison of the performance or effectiveness of different Systems, processes or entities in achieving a specific goal or outcome. It assesses how well one system or entity performs relative to another, often by measuring outputs or outcomes relative to inputs ~~are~~ or resources used. By comparing relative efficiencies, one can identify which option is more efficient and make informed decisions to optimize performance or resource allocation.

Risk Function: OR Loss Function

A risk function, also known as loss function, is a mathematical function that quantifies the potential losses or costs associated with different decisions or actions in a given situation. It is commonly used in theory, statistics and machine learning. The risk function takes into account the uncertainty or variability of outcomes and provides a way to measure and evaluate the potential risks or losses

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Date:

associated with different choices.

In statistical inference, For example, the risk function is used to evaluate the performance of different estim by measuring the expected loss associated with each estimator under different scenarios.

Comparison of asymptotic unbiasedness and consistency.

Asymptotic unbiasedness or consistency are both important concepts in statistical inference particularly estimating the unknown parameters from data. While they are related, they have distinct meanings and implications.

An estimator is asymptotically unbiased if, as the sample size approaches infinity, the expected value of estimator coverages to the true value of parameter being estimated. Asymptotic unbiasedness does not guarantee unbiasedness for finite sample size but indicates that the bias diminshes as the sample size increases.

An estimator is consistent, if the sample size increases, the estimator coverages in probability to the true value of parameter. This means that as more data is gathered,

Date:

the estimators becomes increasingly likely to produce estimates close to the true parameter value.

Consistency ensures that the estimator will converge to the true value over time.

Different forms of Efficiency:

Efficieny in statistics refers to the quality of an estimator in terms of its precision and accuracy. Diff forms of efficieny evaluate the performance of estimators under different criteria.

FORMS:

Date:

① → MSE Efficiency:
It measures the average squared difference b/w the estimator's estimates and the true parameter value. An estimator is considered MSE efficient if it has smallest mean squared error among all unbiased estimators

② → Asymptotic Efficiency:
considers the behaviour of estimators as the sample size increases to infinity. An estimator is considered asymptotically efficient if it achieves the cramer-Rao lower bound (CRLB), which represent the minimum possible variance for an unbiased estimator.

Date:

Write down sufficiency procedure and objection.

To determine whether an estimator is sufficient or not, you can follow the general procedure outlined below:

Step 1: clearly define the est. you want to examine for sufficiency.

Step 2: calculate the likelihood function based on observed data.

Step 3: Factorize the likelihood function. This step involves expressing the likelihood function as the product of two functions

Step 4:

Compare the factorization of the likelihood function with the original estimator.

Date:

Step 5: objections of sufficieny.

1→ **loss of information**

A sufficient statistic compresses the info in data. As a result, it discards some detailed about sample which may be useful.

2→ **Dimension Reduction**

A Suff. st has lower dimensionality than original data. This can be advantagesous for computatoin and storage.

3→ **inadmissibility:**

It is possible for an est. to be sufficient but not admissible.

4→ **Computational complexity:**

Although sufficieny can Simplify the estimation process, it does not guarantee complete efficieny.

Bayes Estimator

Bayes Estimator also known as Bayes Rule, is a concept that allows us to make optimal decisions or predictions based on prior knowledge or observed data. The Bayes Est. is defined as the decision or prediction that minimizes the expected loss ~~of~~ or risk. It combines the prior Dist^n and the likelihood func. to make an optimal decision. The Bayes Est. is

Date:

obtained by minimizing the expected loss with Respect to the posterior Distⁿ.

The Bayes Est. is particularly useful when dealing with uncertain or incomplete information and provide a principled framework for decision making in statistical inference. Calculating the Bayes Est. may involve complex mathematical calculations or numerical approximation, which can be computationally demanding.